

SOLUTION

1. Parallax is a displacement of difference in apparent position of an object viewed along two different lines of sight and it is measured by the angle of semi-angle of inclination between two lines.
3. Resolution of an instrument, is associated to least count error.
5. For visible light range of wavelength is 4000 Å to 7000 Å.
6. The trailing zero(s) in a number with a decimal point are significant.
7. (a) 1AU = average distance between Sun & Earth.
(b) Spherometre is used to measure length as (like screw gauge) less as to 10^{-5} m.
8. All the measurement are upto two places of decimal, least unit is mm so 6.00 mm measurement is most precise.
9. Mass and charge of a particle by using principle of deflection of a charge particle in a magnetic or electric field using spectrometer.
12. Different order of the quantity as velocity of light = 3×10^8 m/s
Speed of person = 10^0 cm/s
Speed of bus = 10^7 cm/hr
13. It is a platinum-iridium alloy cylinder kept at international Bureau of weight and measure at screws near Paris, France.
14. Random error influence precision and systematic error influence the accuracy of result.
15. sin and cos function are dimensionless.
23. Writing the dimensions of either side of the given equation
L.H.S. = S = displacement = $[M^0LT^0]$
R.H.S. = ut = velocity $\times$ time
= $[M^0LT^{-1}][T] = [M^0LT^0]$

and $\frac{1}{2}at^2 = [\text{acceleration}] \times [\text{time}]^2$
= $[M^0LT^{-2}][T]^2 = [M^0LT^0]$
As LHS = RHS
Formula is dimensionally correct.
24. The given equation can be written as,
Pat = b - x²
Now, [Pat] = [b] = [x²]
or [b] = [x²] = $[M^0L^2T^0]$

and $[a] = \frac{[x^2]}{[Pt]} = \frac{[L^2]}{[ML^2T^{-3}][T]} = [M^{-1}L^0T^2]$

25. Let $F = k(m)^x (v)^y (r)^z$... (i)
Here, k is a dimensionless constant of proportionality
Writing the dimensions of RHS and LHS in Eq. (i), we have
 $[MLT^{-2}] = [M]^x [LT^{-1}]^y [L]^z$
= $[M^x L^{y+z} T^{-y}]$
Equating the powers of M, L and T of both sides, we have
 $x = 1, y = 2$ and $y + z = 1$
or $z = 1 - y = -1$
Putting the values in Eq. (i), we get
 $F = kmv^2r^{-1}$

 $F = k \frac{mv^2}{r}$

 $F = \frac{mv^2}{r}$ (where k = 1)
26. a = 6.75×10^3 cm
b = 4.52×10^2 cm
= 0.452×10^3 cm
= 0.45×10^3 cm (upto 2 places of decimal)
 $\therefore a + b = (6.75 \times 10^3 + 0.45 \times 10^3)$ cm
a + b = 7.20×10^3 cm
27. Given l = 21.7 cm
r = 0.46 mm = 0.046 cm
volume of wire
 $V = \pi r^2 l$

= $\frac{22}{7} (0.046)^2 (21.7)$
= $0.1443 \text{ cm}^3 = 0.14 \text{ cm}^3$
28. (a) Least count of vernier callipers
= 1 MSD - 1 VSD

= 1 MSD - $\frac{19}{20}$ MSD = $\frac{1}{20}$ MSD

= $\frac{1}{20}$ mm = $\frac{1}{200}$ cm = 0.005 cm
(b) Least count of screw gauge

= $\frac{\text{Pitch}}{\text{Number of divisions of circular scale}}$

= $\frac{1}{100}$ mm = $\frac{1}{1000}$ cm = 0.001 cm
(c) Wavelength of light, $\lambda = 10^{-5}$ cm = 0.00001 cm
 $\therefore$ Least count of optical instrument = 0.00001 cm
Thus, clearly the optical instrument is the most precise.